MIFECO

MARS HARNESS LAB / VERSION 5

FROM RESEARCH IDEATION TO VERIFIED ACTION - FASTER

A practical intelligence layer for science, engineering, and mission delivery.

(A lot) More mission learning per hour of expert attention.



15 MIN TALK / 5 MIN DISCUSSION

MOXIE shows than we can produce O^2 on Mars.

As a demonstration we asked the Ai advanced harnesses – Can we scale this to industrial levels to refuel a start ship in 2 years.

OBSERVED

12 g/h

At its most efficient, the flight unit produced oxygen on Mars.

Reported by NASA • a measured demonstration result

WHAT WAS MODELED

SCALED to 34kg/h

600 tons – 3+ orders of magnitude

Massive volume of O2 is a requirement to make habitats and the ability to return people to Mars possible. This is one path to do that.

Design analysis • assumptions, simulation and models, and designs are needed

THE RESULT

TO DEMONSTRATE

The Ai harnesses answered this via a scientific paper with analysis, a engineering design for a prototype and a operations plan for Plant-level reliability, maintenance, and mission performance. And yes the AI says it is possible

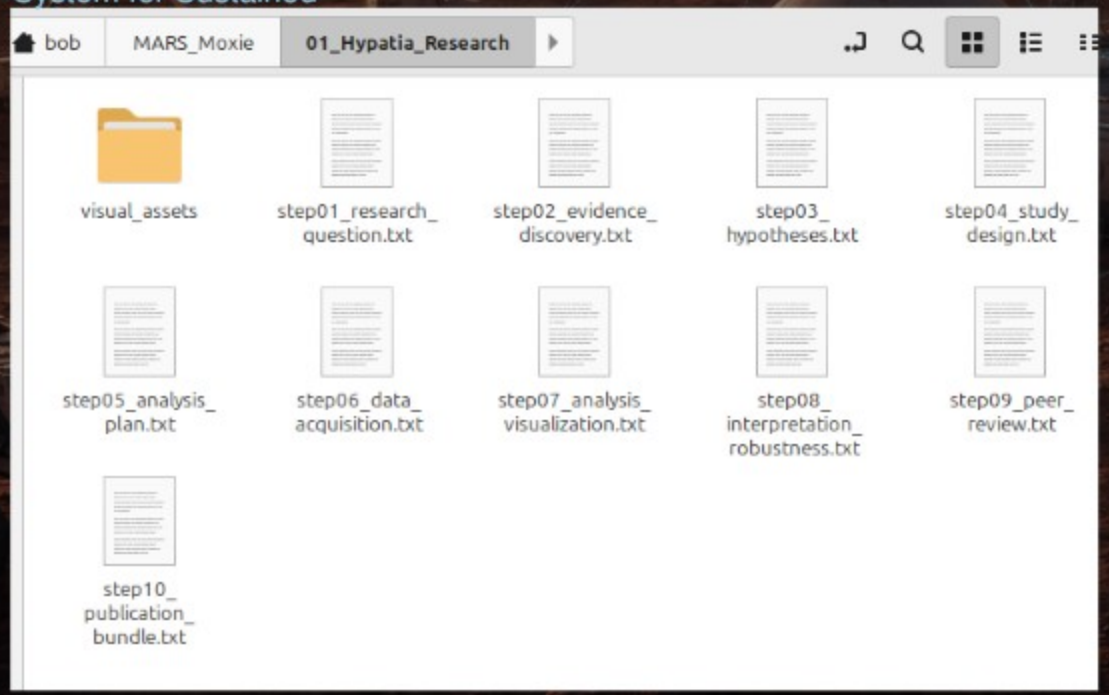
Tests and are still required

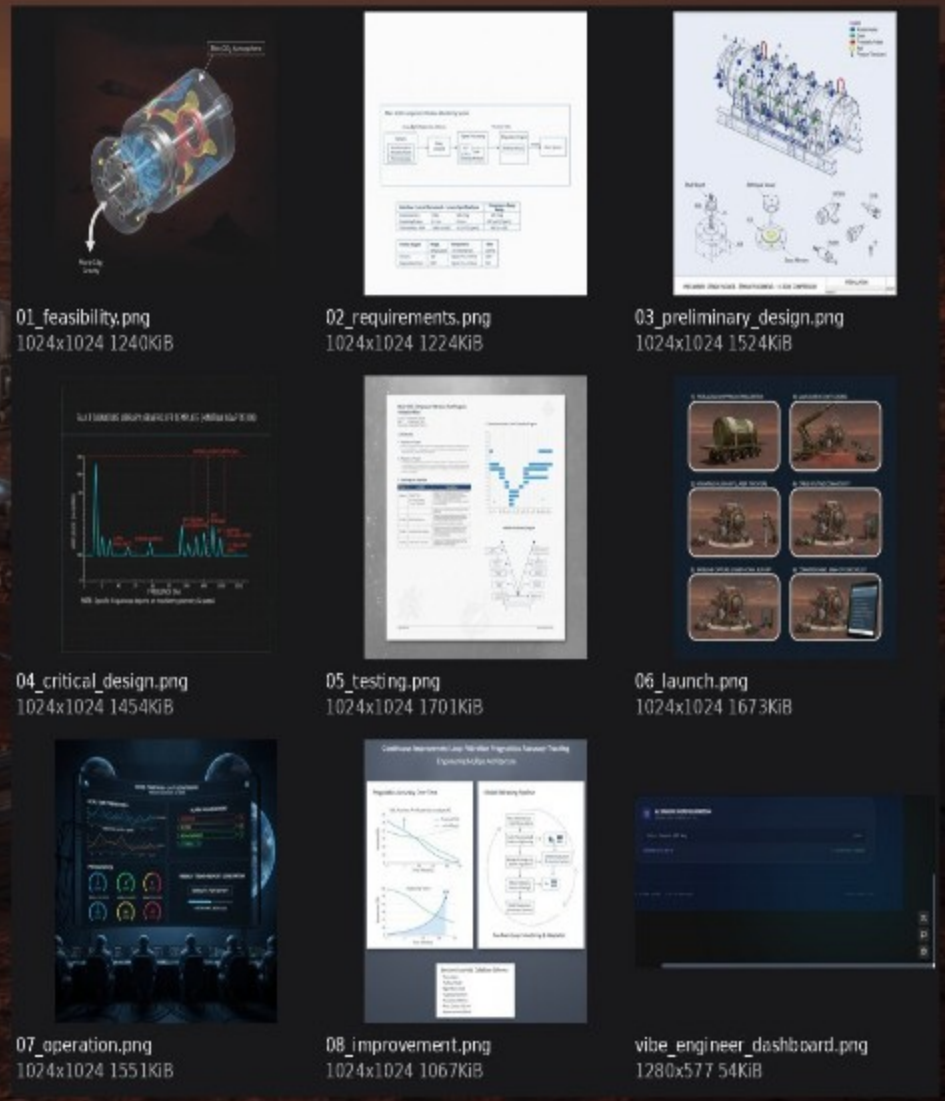
The result is a clearer path from evidence to the next test scaling the process. This will save thousands of hours of work, not all human work, but much of the documentation, providing the researchers and engineers with rapid ideation and a complete framework to build on.

The case study result – 145 artifacts - and yes, it is possible

Doing the research... It is possible to scale MOXIE?

Step 10: Publication Bundle - Project ARES-SOE
Title: Industrial-Scale Solid Oxide CO_2 Electrolysis (SOXE) for Human Mars Missions: A 33.8 kg/hr Modular Architecture for Habitat Life Support and Starship Propellant Refueling
Author List: Hypatia AI (Lead Systems Architect), Mars In
1. Abstract As human Mars exploration transitions from exploration to habitation, In-Situ Resource Utilization (ISRU) must scale by three orders of magnitude. This paper presents the design and validation framework for Project ARES-SOE (Advanced Regenerative Electrolysis System for Sustained Oxygen Production), a scaled-up solid-oxide CO_2 electrolysis system. Building on the heritage of the NASA MOXIE instrument, the system is designed to produce 592.4 t of O_2 over 2 Earth years (17,520 hours)—sufficient to sustain a 12-person habitat and provide 50% of





First principles reveals better design options

1 ASSUMPTIONS

Law, measured fact, estimate, habit or opinion

2 FUNDAMENTALS

Dated sources, units, bounds and uncertainty

3 COST GAP

Current baseline versus a feasible lower bound

4 REBUILD

Bold redesign and conservative alternative

5 STRESS TEST

First failure, skeptical challenge and test

6 DECISION

Best supported option and first reversible step

Deliver assumptions, source ledger, cost-gap table and a 30-day plan.

Every action returns a checkable receipt

CONTRACT

Goal, inputs and acceptance

ACTION

One bounded tool step

RECEIPT

Result, paths and error

VERIFY

Evidence against criterion

PASS

Evidence satisfies the check

FAIL

Evidence contradicts it

BLOCKED

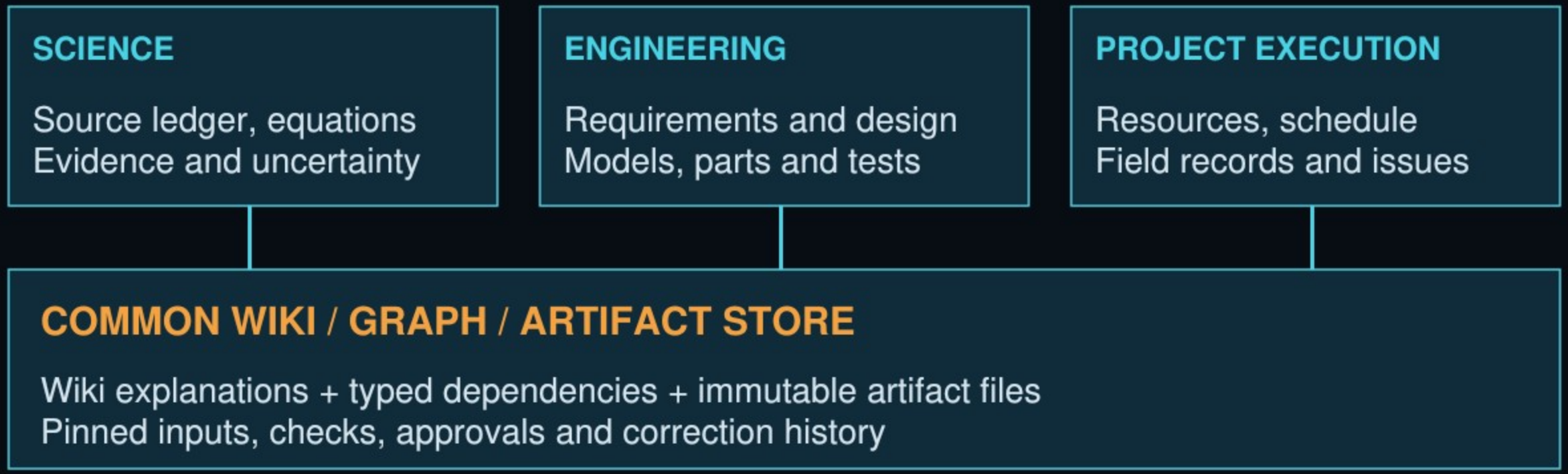
A dependency prevents it

UNVERIFIED

Evidence is insufficient

Repair within limits. Save a checkpoint. Deliver artifacts with evidence.

Specialized harnesses share a common artifact graph



Approved outputs flow forward. Targeted correction requests can return upstream.

Downstream blockers trigger controlled regeneration

DETECT	REQUEST	REGENERATE	REVALIDATE
Failed check or execution blocker	Pinned revisions and bounded question	Upstream owner creates a revision	Rebuild dependents and check release

B1 stays in history. Correction branch C1 holds revised work.

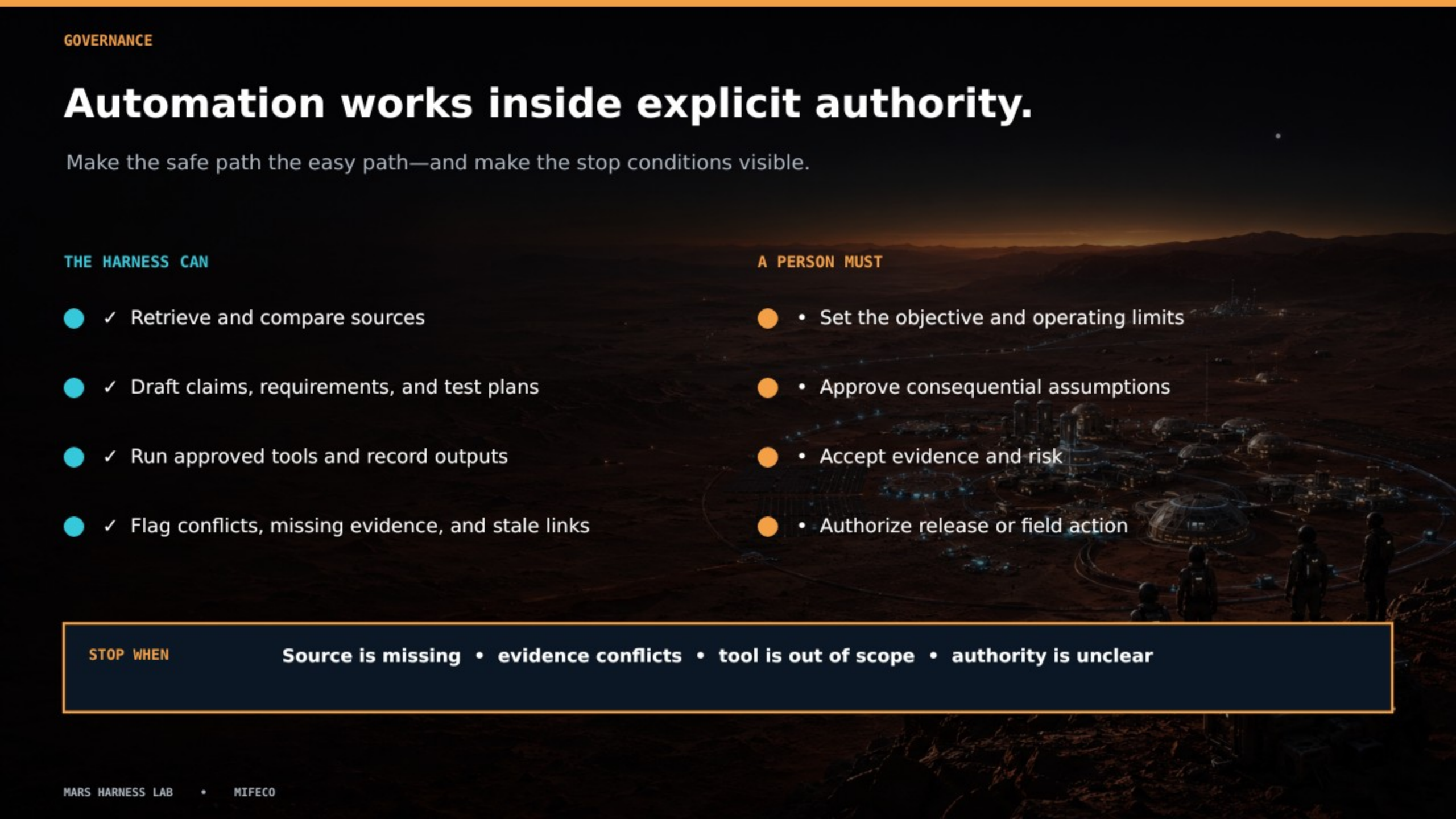
B2 becomes current only after approved upstream evidence, downstream checks and required authority agree.

Stale inputs block promotion. Unrelated work needs a recorded independence check.

GOVERNANCE

Automation works inside explicit authority.

Make the safe path the easy path—and make the stop conditions visible.



THE HARNESS CAN

- ✓ Retrieve and compare sources
- ✓ Draft claims, requirements, and test plans
- ✓ Run approved tools and record outputs
- ✓ Flag conflicts, missing evidence, and stale links

A PERSON MUST

- • Set the objective and operating limits
- • Approve consequential assumptions
- • Accept evidence and risk
- • Authorize release or field action

STOP WHEN **Source is missing • evidence conflicts • tool is out of scope • authority is unclear**

Reviewed lessons improve the next working loop

tasks/todo.md

Goal and bounded plan
Observable acceptance checks
Progress and final review

tasks/lessons.md

Mistake and root cause
Scoped prevention rule
Early detection signal

Resume from checked work. Re-plan after evidence changes. Keep workers bounded.

97 requirements mapped. Local checks support the methods. Live host enforcement still needs evidence.

GET STARTED

Get started in three ways.

Free lab downloads, paid preconfigured applications, and open-source references.

FREE LAB MATERIALS

Apache License 2.0

Instructions, skills, templates, and example workflows. These can run on most agentic harnesses', Grok Bot, Hermes, Open Claw, Claude Code, Codex, etc.

OPEN DOWNLOADS ↗

This contains free downloads for agents, skills, code, memory managment, etc.

PRECONFIGURED ACCESS

MIFECO applications

Hosted versions for science, engineering, and project execution.

These are paid version to cover the Ai token expenses as these use frontier models extensively

VIEW MIFECO CATALOG ↗

Project Hypatia Pro ↗

Vibe Engineer ↗

PM Accelerator ↗

OPEN SOURCE

Robertstar2000 on GitHub

Review source repositories behind the specialist workflows.

EXPLORE GITHUB ↗

Hypatia-Pro repo ↗

Intelligent-Engineer-Pro repo ↗

project-management-accelerator ↗

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DISCUSSION

Questions & conversation

What would you want to put through one complete loop?

5 MINUTES